Zohaib Aftab, Ph.D.

Claude Certified Architect

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Leadership & Governance: 10+ years of experience driving enterprise digital transformation, managing cross-functional technical teams using Scrum methodologies, and establishing rigorous MLOps, Data Governance, and SDLC.
AI & Agentic Workflows: Expert in designing production-grade multi-agent systems, RAG pipelines, and secure Python microservices integrated into complex enterprise workflows.
Data & Cloud: Deep experience processing data workloads using Apache Spark, Databricks, and containerized microservices on Azure Kubernetes Service (AKS).
Communication: Actively mentoring data engineers and collaborating across global sprint teams in both English and French.

RELEVANT EXPERIENCE

Specialist - AI | Synechron **09/2024 – Present**

Direct the end-to-end delivery of enterprise AI products as a solutions engineer, acting as the Sr. AI Engineer/Architect for client deliverables (US Bank), system design, and data governance.

- Project delivery in Wealth and Asset Management:
  - Rolled out production agentic systems at US Bank that integrates information from SQL, Swagger, Confluence, and SharePoint into **RAG** pipelines to provide accurate, grounded answers using an NLQ interface. Adopted by 600 users, reducing API discovery time by 90% and saving over **$1M** in annual operational efficiency gains.
  - Delivered **multi-agent systems** using LangGraph integrating complex workflows across enterprise platforms including Salesforce, SharePoint, and Confluence streamlining account operations. Built evaluation pipelines to measure performance
  - Architected enterprise-grade **ServiceNow** incident resolution platform. Defined strict service boundaries and designed reusable abstract base classes and robust data validation schemas, allowing disparate engineering teams to seamlessly connect their own data sources to the core orchestration engine.
- Mentorship & Enablement:
  - Mentor junior engineers and establish engineering discipline, testing playbooks and reference architectures for AI-enabled coding with guardrails.
- Business development:
  - Support pre-sales activities, including solution architecture and tailored technical proposals for prospective customers.

Tech Stack: FastAPI, LangGraph, SQL, Docker, Github Actions, GitLab pipelines, WebSocket APIs, Langfuse (observability), Azure Kubernetes Service (AKS), MS and Claude Agent SDKs, React, MCP

Senior Data Scientist **10/2023 – 09/2024**
B-yond | *Montreal, Canada*

At this startup, I engineered and scaled distributed data pipelines using Apache Spark and **Databricks** on

Microsoft **Azure** to efficiently process massive volumes of Wi-Fi network performance data.

- **Big data analysis for Wi-Fi network data**: Built end-to-end time-series analysis models on top of these **Databricks pipelines** to identify network quality of service (QoS) trends and operational patterns.
- **LLM-powered smart chatbot**: Developed a low-latency information retrieval system (**RAG**) from diverse sources including a large corpus of technical documentation (50k+ pages) and a **Postgres SQL** database, achieving precision and recall of over 90%. Collaborated with cross-functional teams, including architecture and frontend teams, to serve models the tool to internal clients. Link to blog
- **Topic modeling** for JIRA recommender system: Developed topic models using LDA and BERTopic to identify key themes in JIRA tickets for a major European telecom client (Orange). This adaptation pinpointed essential themes in JIRA tickets, dramatically boosting the system's ability to suggest resolution path and streamlining issue resolution efforts across multiple departments
- Collaborated with architecture teams to serve these AI models to internal clients, implementing robust **MLOps** practices, including CI/CD pipelines and model tracking via **MLflow**—to ensure reliable deployment.

Research Data Scientist **07/2018 – 06/2023**
National Center of Robotics and Automation | *Pakistan*

Directed cross-functional engineering and research teams to execute large-scale government projects with budgets exceeding $200,000, successfully delivering 2 commercial AI products and securing 3 patents.

- Architected robust data pipelines to ingest, process, and manipulate over 8 million raw data points (11+ hours of sequential data), establishing the foundational data infrastructure for high-accuracy predictive models.
- Guided the technical strategy for deep learning initiatives, overseeing the team's implementation of 1D temporal CNN architectures in TensorFlow for complex human activity recognition systems.

Lead Data Scientist **03/2023 – 09/2023**
Senarios | *Canada [Remote]*

- Directly managed and mentored a team of 5 Machine Learning Engineers, overseeing the end-to-end architecture and delivery of enterprise AI products for major US corporate clients.
- Acted as the primary technical liaison for executive stakeholders, translating business requirements into scalable machine learning pipelines and actionable technical roadmaps.

EDUCATION & CERTIFICATIONS

Ph.D. in Robotics **2012**
Université de Lyon| *France*
Formulated multivariable optimization problem with inequality constraints for simulation of humanoid robots' balance recovery. Implemented using Quadratic Programming (QP) and with MPC.

Masters in Automation **2009**
Ecole Centrale Nantes| *France*

Claude Certified Architect - Foundations **2026**
Anthropic | Proctored exam
Validates hands-on experience building production-grade multi-agent systems, MCP tool integration, and Claude Code workflows real-world enterprise use cases. Link to batch